

# TAI Exoplanet Unit: Teacher Overview

*Confidential teacher reference. Contains the chemistry real/decoy answers; not for students.*

Each of the eight groups works on one exoplanet for the whole week, across three subjects. This overview gives, per planet, the information each subject needs, plus a link to where the real published data can be found. Physics and maths use the transit light curves to recover the planet's radius, orbital period and semi-major axis; chemistry runs the rover 'brine' sample (a real iron(III) chloride solution for five groups, a look-alike decoy for three); biology designs an organism for the planet's main survival challenge.

## At a glance

| Group | Planet       | Host star (type)  | Distance | Chemistry sample | Biology: survival challenge                                            |
|-------|--------------|-------------------|----------|------------------|------------------------------------------------------------------------|
| 1     | 55 Cancri e  | 55 Cancri A (G8V) | 41 ly    | REAL ~3.2 M      | Extreme heat and volcanic toxicity                                     |
| 2     | WASP-76 b    | WASP-76 (F7V)     | 634 ly   | Decoy (food dye) | Aerial life in gas-giant clouds with huge day-night temperature swings |
| 3     | TRAPPIST-1 e | TRAPPIST-1 (M8V)  | 41 ly    | REAL ~1.6 M      | Intense stellar flares and radiation from an active red dwarf          |
| 4     | K2-18 b      | K2-18 (M2.5V)     | 124 ly   | Decoy (food dye) | No sunlight in a deep ocean beneath a thick hydrogen atmosphere        |
| 5     | TOI-700 d    | TOI-700 (M2.5V)   | 102 ly   | REAL ~0.8 M      | Tidal locking: a permanent day side and night side                     |
| 6     | GJ 1214 b    | GJ 1214 (M4V)     | 48 ly    | Decoy (food dye) | Crushing atmospheric pressure and permanent haze                       |
| 7     | LHS 1140 b   | LHS 1140 (M4.5V)  | 49 ly    | REAL ~0.2 M      | Dim red-dwarf starlight and high surface gravity                       |
| 8     | TRAPPIST-1 g | TRAPPIST-1 (M8V)  | 41 ly    | REAL ~0.1 M      | Permanent extreme cold, possibly beneath a global ice layer            |

## The eight planets

### Group 1: 55 Cancri e (55 Cancri A, G8V, 41 ly)

*Ultra-hot rocky super-Earth with a permanent global lava ocean; JWST confirmed a real CO/CO<sub>2</sub> atmosphere (Hu et al. 2024).*

## Physics & maths

|                                |                                             |
|--------------------------------|---------------------------------------------|
| Host star                      | 55 Cancri A (G8V), 41 ly                    |
| Star mass / radius / temp.     | 0.905 M $\odot$ / 0.943 R $\odot$ / 5172 K  |
| Planet radius / mass / gravity | 1.88 R $\oplus$ / 7.99 M $\oplus$ / ~2.27 g |
| Orbital period                 | 0.7365 d                                    |
| Semi-major axis                | 0.01544 AU                                  |
| Transit depth / duration       | 331 ppm (0.033%) / 1.7 h                    |

**Chemistry.** REAL sample: iron(III) chloride, most concentrated (~3.2 M). The hottest, most evaporative world, framed as the richest brine, so it should give the largest iron-oxide yield of the five real groups.

**Biology.** Extreme heat and volcanic toxicity. Favours heat-stable proteins and enzymes, always-on heat-shock repair, and a metabolism based on volcanic chemistry (sulfur, CO<sub>2</sub>) rather than sunlight. Earth analog: *Pyrolobus fumarii*, a vent archaeon that grows to 113 C (Blochl et al. 1997).

**Real data.** [NASA Exoplanet Archive overview: 55 Cancri e](#)

## Group 2: WASP-76 b (WASP-76, F7V, 634 ly)

*Ultra-hot gas giant (an 'ultra-hot Jupiter'); ESPRESSO detected atomic iron vapour, the famous 'iron rain' (Ehrenreich et al. 2020). No solid surface.*

## Physics & maths

|                                |                                                                                                  |
|--------------------------------|--------------------------------------------------------------------------------------------------|
| Host star                      | WASP-76 (F7V), 634 ly                                                                            |
| Star mass / radius / temp.     | 1.458 M $\odot$ / 1.756 R $\odot$ / 6329 K                                                       |
| Planet radius / mass / gravity | 20.78 R $\oplus$ (1.85 R <sub>Jup</sub> ) / 284.14 M $\oplus$ (0.89 M <sub>Jup</sub> ) / ~0.66 g |
| Orbital period                 | 1.81 d                                                                                           |
| Semi-major axis                | 0.033 AU                                                                                         |
| Transit depth / duration       | 1.17% / 3.8 h                                                                                    |

**Chemistry.** DECOY: a food-colourant solution in a matching amber hue, with no real iron. A gas giant with no solid surface and no evaporite brine, so testing produces no iron-oxide precipitate. Discovering the sample is a decoy is itself a valid, interesting result.

**Biology.** Aerial life in gas-giant clouds with huge day-night temperature swings. With no surface to stand on, favours gas-filled floats for buoyancy and drifting between the hot dayside and cooler nightside. One of the two most speculative cases: the nearest analog is the classic 'floater' organism proposed for Jupiter's clouds (Sagan and Salpeter 1976), now studied for Venus.

**Real data.** [NASA Exoplanet Archive overview: WASP-76 b](#)

## Group 3: TRAPPIST-1 e (TRAPPIST-1, M8V, 41 ly)

*Roughly Earth-sized rocky planet in the habitable zone of the TRAPPIST-1 ultracool dwarf; Earth-like density. No atmosphere confirmed.*

## Physics & maths

|                                |                                             |
|--------------------------------|---------------------------------------------|
| Host star                      | TRAPPIST-1 (M8V), 41 ly                     |
| Star mass / radius / temp.     | 0.090 M $\odot$ / 0.119 R $\odot$ / 2566 K  |
| Planet radius / mass / gravity | 0.92 R $\oplus$ / 0.69 M $\oplus$ / ~0.82 g |
| Orbital period                 | 6.101 d                                     |
| Semi-major axis                | 0.02925 AU                                  |
| Transit depth / duration       | 0.50% / 0.9 h                               |

**Chemistry.** REAL sample: iron(III) chloride, ~1.6 M (upper-middle of the gradient).

**Biology.** Intense stellar flares and radiation from an active red dwarf. Favours very fast DNA repair, protective (or radiation-using) pigments, and the ability to shut down and revive after a flare. Earth analog: *Deinococcus radiodurans*, which repairs shattered DNA (Cox and Battista 2005); flare UV at TRAPPIST-1 modelled by Estrela et al. 2020.

**Real data.** [NASA Exoplanet Archive overview: TRAPPIST-1 e](#)

#### Group 4: K2-18 b (K2-18, M2.5V, 124 ly)

*Sub-Neptune in the habitable zone; JWST detected methane and a disputed dimethyl-sulfide 'biosignature' claim (Madhusudhan et al. 2023). A good case study in scientific caution.*

##### Physics & maths

|                                |                                             |
|--------------------------------|---------------------------------------------|
| Host star                      | K2-18 (M2.5V), 124 ly                       |
| Star mass / radius / temp.     | 0.359 M $\odot$ / 0.411 R $\odot$ / 3457 K  |
| Planet radius / mass / gravity | 2.37 R $\oplus$ / 8.92 M $\oplus$ / ~1.59 g |
| Orbital period                 | 32.94 d                                     |
| Semi-major axis                | 0.1429 AU                                   |
| Transit depth / duration       | 0.28% / 3.5 h                               |

**Chemistry.** DECOY: a food-colourant solution in a matching amber hue, with no real iron. A sub-Neptune with no solid surface, so testing produces no iron-oxide precipitate.

**Biology.** No sunlight in a deep ocean beneath a thick hydrogen atmosphere. Favours chemosynthesis (energy from dissolved hydrogen, methane or sulfur compounds) and symbiosis with chemosynthetic microbes. Earth analog: hydrothermal-vent ecosystems that live without sunlight (Corliss et al. 1979); Hycean-world energetics modelled by Madhusudhan et al. 2023.

**Real data.** [NASA Exoplanet Archive overview: K2-18 b](#)

#### Group 5: TOI-700 d (TOI-700, M2.5V, 102 ly)

*The first Earth-sized habitable-zone planet found by TESS (2020). Tidally locked; no atmosphere detected yet, so published climate pictures are models.*

##### Physics & maths

|                                |                                             |
|--------------------------------|---------------------------------------------|
| Host star                      | TOI-700 (M2.5V), 102 ly                     |
| Star mass / radius / temp.     | 0.415 M $\odot$ / 0.421 R $\odot$ / 3459 K  |
| Planet radius / mass / gravity | 1.07 R $\oplus$ / 1.25 M $\oplus$ / ~1.09 g |
| Orbital period                 | 37.42 d                                     |
| Semi-major axis                | 0.1633 AU                                   |
| Transit depth / duration       | 544 ppm (0.054%) / 3.5 h                    |

**Chemistry.** REAL sample: iron(III) chloride, ~0.8 M (lower-middle of the gradient).

**Biology.** Tidal locking: a permanent day side and night side. Favours life in the twilight 'terminator' band, large low-light eyes, and tolerance of a steep temperature gradient. Earth analog is theoretical: climate models of tidally-locked 'eyeball' worlds (Yang et al. 2013; Pierrehumbert 2011).

**Real data.** [NASA Exoplanet Archive overview: TOI-700 d](#)

### Group 6: GJ 1214 b (GJ 1214, M4V, 48 ly)

*Sub-Neptune with a thick, high-metallicity haze; the benchmark 'featureless spectrum' world (Kempton et al. 2023). No solid surface.*

#### Physics & maths

|                                |                                             |
|--------------------------------|---------------------------------------------|
| Host star                      | GJ 1214 (M4V), 48 ly                        |
| Star mass / radius / temp.     | 0.182 M $\odot$ / 0.216 R $\odot$ / 3101 K  |
| Planet radius / mass / gravity | 2.73 R $\oplus$ / 8.41 M $\oplus$ / ~1.13 g |
| Orbital period                 | 1.58 d                                      |
| Semi-major axis                | 0.01505 AU                                  |
| Transit depth / duration       | 1.34% / 0.9 h                               |

**Chemistry.** DECOY: a food-colourant solution in a matching amber hue, with no real iron. A sub-Neptune with no solid surface, so testing produces no iron-oxide precipitate.

**Biology.** Crushing atmospheric pressure and permanent haze. Favours pressure-tolerant membranes and enzymes (piezolytes) and chemistry-based (not light-based) energy, since haze blocks sunlight. Earth analog: Mariana-Trench microbes found down to about 10,900 m (Yayanos 1995; Bartlett 2002).

**Real data.** [NASA Exoplanet Archive overview: GJ 1214 b](#)

### Group 7: LHS 1140 b (LHS 1140, M4.5V, 49 ly)

*Dense rocky super-Earth / water-world candidate; high gravity and a quiet host star make it one of the best targets to search for a rocky-planet atmosphere (Cadieux et al. 2024).*

#### Physics & maths

|                                |                                             |
|--------------------------------|---------------------------------------------|
| Host star                      | LHS 1140 (M4.5V), 49 ly                     |
| Star mass / radius / temp.     | 0.184 M $\odot$ / 0.216 R $\odot$ / 3096 K  |
| Planet radius / mass / gravity | 1.73 R $\oplus$ / 5.60 M $\oplus$ / ~1.87 g |
| Orbital period                 | 24.74 d                                     |
| Semi-major axis                | 0.0946 AU                                   |
| Transit depth / duration       | 0.54% / 2.2 h                               |

**Chemistry.** REAL sample: iron(III) chloride, ~0.2 M (near the dilute end of the gradient).

**Biology.** Dim red-dwarf starlight and high surface gravity. Favours far-red / infrared photosynthetic pigments and compact, robust body structures for elevated gravity. The high-gravity half is the more speculative. Earth analog: chlorophyll f for far-red light (Chen et al. 2010); plant hypergravity studies (Hosamani et al. 2023).

**Real data.** [NASA Exoplanet Archive overview: LHS 1140 b](#)

### Group 8: TRAPPIST-1 g (TRAPPIST-1, M8V, 41 ly)

*Cold rocky planet at the outer edge of the TRAPPIST-1 habitable zone; near-Earth gravity and density. Pairs with group 3 for a same-star, different-orbit comparison. No atmosphere confirmed.*

#### Physics & maths

|                                |                                             |
|--------------------------------|---------------------------------------------|
| Host star                      | TRAPPIST-1 (M8V), 41 ly                     |
| Star mass / radius / temp.     | 0.090 M $\odot$ / 0.119 R $\odot$ / 2566 K  |
| Planet radius / mass / gravity | 1.13 R $\oplus$ / 1.32 M $\oplus$ / ~1.04 g |
| Orbital period                 | 12.35 d                                     |
| Semi-major axis                | 0.04683 AU                                  |

|                                 |               |
|---------------------------------|---------------|
| <b>Transit depth / duration</b> | 0.75% / 1.2 h |
|---------------------------------|---------------|

**Chemistry.** REAL sample: iron(III) chloride, most dilute (~0.1 M). The coldest world, framed as the weakest brine, so it should give the smallest iron-oxide yield of the real groups.

**Biology.** Permanent extreme cold, possibly beneath a global ice layer. Favours antifreeze molecules and cold-active enzymes and membranes that stay flexible near freezing. Earth analog: antifreeze glycoproteins in Antarctic fish (DeVries 1971); life in a lake sealed under the Antarctic ice sheet (Christner et al. 2014).

**Real data.** [NASA Exoplanet Archive overview: TRAPPIST-1 g](#)

## Running the chemistry comparison (real vs decoy)

Five groups get a genuine iron(III) chloride solution and three get a food-colourant decoy in a matching amber hue. Label every sample by planet name only, never 'real' or 'fake'; students find out by testing (a base added to a real sample gives a red-brown iron-oxide/hydroxide precipitate; the decoy does not). The five real concentrations form a gradient, deliberately mapped to each planet's temperature and evaporation, so the recovered iron-oxide mass should roughly follow that ranking. Only the relative order matters, so you can adjust the molarities to whatever stock you have. At the final conference, groups compare yields against the temperature-based prediction, and the decoy groups report that finding too.

## Transit graphs and data files

The transit light curves in the student decks are simulated from each planet's real parameters (radius, star radius, period), so every value students measure comes out true while difficulty stays uniform across groups. Each group's raw single-transit points are in `Transit_data\GroupN_..._transit.csv`, and `plot_transit_data.py` regenerates the plots. True values are in `Teacher_Transit_answer_key.csv` and annotated answer-key graphs in `Teacher_TransitGraphs_KEY.pptx`. Group 2's deck also has a slide comparing a real TESS transit of WASP-76 b with our simulation (real points in `Transit_data\WASP76b_REAL_TESS_transit.csv`).

## Checking the numbers

All values in the planet tables are published measurements, cross-checked against the NASA Exoplanet Archive. They are also mutually consistent: each planet's semi-major axis follows from its orbital period and host-star mass by Kepler's third law, and its transit depth follows from the planet and star radii, which is exactly what students reconstruct. To check any value yourself, open that planet's link in the section above and look at its 'Planet Parameters' listing, which shows every published value with its source paper. Exoplanet parameters legitimately vary between studies, so a specific paper may differ from the value here by 10 to 20 percent (planet masses in general, and the radius and host-star mass of K2-18 b and TOI-700 d, carry the most spread); the same figures are used consistently across these tables, the transit graphs and the answer key.

## Finding the real data (European sources)

The per-planet links go to the NASA Exoplanet Archive, the canonical place the real measured parameters, references and light-curve data are collected. For European (ESA-led) sources, useful starting points are:

- CDS VizieR (Strasbourg): published light-curve tables, e.g. TRAPPIST-1. <https://vizier.cds.unistra.fr>

- ESA CHEOPS photometry via the SSDC mirror (55 Cnc e, WASP-76 b, GJ 1214 b, LHS 1140 b). <https://www.ssdsc.asi.it/cheops/>
- ESA Ariel / ExoClock: education-friendly real transit light curves. <https://www.exoclock.space>
- Exoplanet Transit Database (VarAstro, Czech Astronomical Society). <http://var2.astro.cz/ETD/>